University of Karachi
Department of Public Administration
BSPA 2nd Year Basic Mathematics – 407

Note: This assignment must be **completed by hand**. Typed or printed submissions will not be accepted. Copying or sharing of work will be considered **academic dishonesty**. This assignment must be submitted by **Monday, 27 April 2026**. Late submissions will not be accepted.

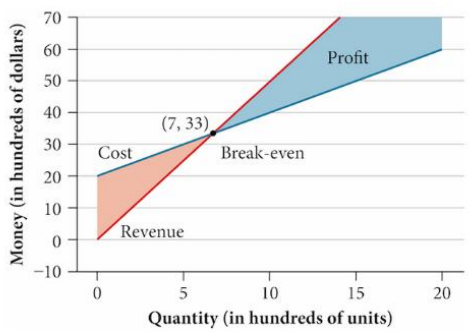
TOPIC: INVESTIGATING PROFITS VIA SYSTEMS OF EQUATIONS

Question 1:

a. How you can use the system of linear equation to investigate profit give two applications?

b. Interpret/Analyze figure. a

Figure. a



c. Explain the following

- Breakeven point
- Cost Function
- Profit Function
- Revenue Function

d. Can a system of linear equations have exactly two solutions? Explain why or why not.

e. If you are performing a break-even analysis for a business and their cost and revenue equations are dependent, explain what this means for the company's profit margins.

f. If you are solving a break even analysis and get a negative break-even point, explain what this signifies for the company?

g. If you are solving a break even analysis and there is no break-even point, explain what this means for the company. How should they ensure there is a break-even point

Question 2:

a. A publishing company produces academic workbooks for university students. The company incurs a fixed weekly cost of $12,000 for printing setup, staff salaries, and administrative expenses. Additionally, the variable cost per workbook is $8, which includes paper, ink, and binding. Each workbook is sold at a price of $20. Find

1. Cost Function
2. Revenue Function
3. Break-Even Point
4. Profit Function

5. Calculate the profit or loss when the company produces and sells 1,500 workbooks in a week. State clearly whether the company is operating at a profit or a loss.

6. Interpret you results. Briefly explain how changes in the selling price or production cost would affect the break-even point.

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b. The startup cost for a restaurant is $120,000, and each meal costs $10 for the restaurant to make. If each meal is then sold for $15, after how many meals does the restaurant break even?